Exercise 2: Interrogating the data

In this exercise we will review the database for IndoTERM. A good understanding of the structure of the initial database is fundamental. The initial database contains information regarding the structure of the Indonesian economy, especially the regional characteristics of the economy. It also forms an initial solution of the IndoTERM model equations.

1. Getting Started

You will be working in the working directory of **C:\DTERMCRS\EX2.** The aim of this exercise is to have a closer look at the initial data file for IndoTERM. Work through this exercise and write down answers to the questions onto the instruction sheet. The questions are scattered through the instructions.

2. Viewing the data using ViewHAR

You can view the data either directly in ViewHAR or by typing commands in DOS.

2.1 Opening ViewHAR directly

You can open ViewHAR directly by clicking on the ViewHar icon on your desktop. In the toolbar click on **File | Open HAR or SL4 file** and navigate to the working directory **C:\ DTERMCRS \ EX2.** This directory contains two *.HAR files (aggmod.har and aggsets.har). Click on aggmod.har.

2.2 ViewHAR via DOS commands

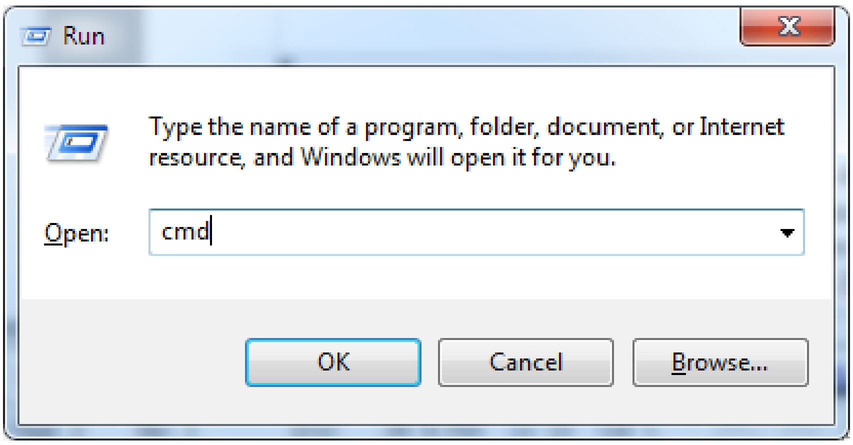
2.2.1 Opening a DOS box

If you already know how to open a DOS box, do it now, and jump to the next subsection, *Navigating to the work folder*.

There are many ways to open a DOS box -- but some only work with some versions of Windows. Here is one method that works with with most versions.

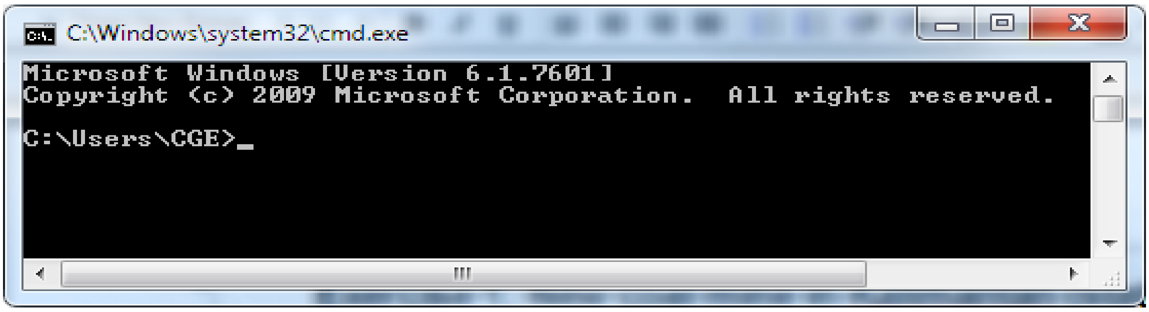
Locate the Windows key near bottom left of your keyboard. It may look like this  or this .

Simultaneously press the Windows key and the "R" key. The Run box should be displayed:



Type in "cmd", then click OK.

You should see a DOS box rather like this:

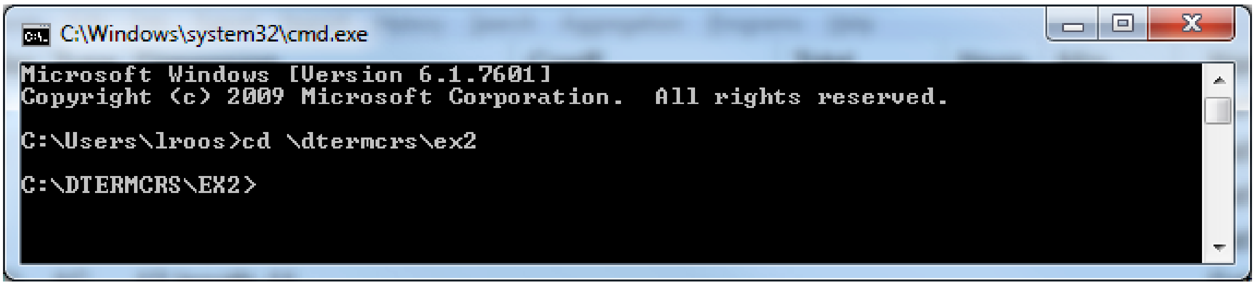


2.2.2 Navigating to the work folder

Type the following line, exactly as shown below, into your DOS box, then press Enter:

```
cd \DTERMCRS\Ex2
```

The DOS box should now resemble:



Now type:

```
dir
```

to list the files in this folder.

To view the data file type:

```
viewhar aggmod.har
```

This is the main data file for IndoTERM -- it contains Indonesian data for the year 2005. Use this file to answer the questions below.

3. Questions

3.1. Which Header contains the set of industries? *IND*
How many commodities are there, and how many industries? *41 COM, 41 IND*

3.2. How many margins commodities are there? *6*
Which commodities are they?

Trade, RailTrans, RoadTrans, WaterTrans, AirTrans, TransSvc

3.3. What does the Header "TRAD" contain?

TRADE shows the value of inter-regional trade by sources (r in ORG) and destinations (d in DST) for each good (c in COM) whether domestic or imported (s in SRC). The diagonal of this matrix (r=d) shows the value of local usage which is sourced locally. For foreign goods (s="imp")the regional source subscript r (in ORG) denotes the port of entry.

How many dimensions does this Header have? What are they?

4 dimensions, namely COM, SRC, ORG, DST

3.4. What is the name of the set containing the regions of origin? *ORG*
What are the regions of origin?

WestSumatra, Eastsumatra, NthWestJava, EastJava, WKalimantan, EKalimantan, NthSulawesi, SthSulawesi, Bali, NusaTeng, Maluku, Papua

3.5. In the top right hand corner of the Trade matrix (Header TRAD) there are four drop-down windows. Click on them until you have “Sum COM”, “Sum SRC”, “All ORG” and “All DST”. Describe the matrix that you see in front of you.

This matrix shows the trade between regions. The diagonal values of the matrix is the largest, implying that most goods are sourced locally.

What is the share of all commodities (domestic and imported) used in East Java are sourced from East Java? [Hint: use column shares]

76%

Now change the four settings to “FoodProds”, “1 dom”, “All ORG” and “All DST” and explain again what you have in front of you. (Note: ViewHAR can only view two dimensions at a time. See the ViewHAR help under "Options for real matrices" for more details).

The values show the trade of domestically sourced FoodProds between regions in Indonesia.

Set the data to row shares. What do you see?

Most of the domestically source FoodProds are sourced from the local region. This matrix is strongly diagonal.

3.6. What is the name of the COEFFICIENT used in TERM.TAB to represent the "Armington elasticities", ie, the elasticities of substitution between imported and Indonesian goods?

SIGMADOMIMP

From what Header are these values read? *P015*

What are the values used for these Armington elasticities? What do these parameters represent?

The ease with which we can substitute between domestically and imported commodities due to changes in relative price

Which products have the highest elasticities?

OilGas 5.209, Paddy 5.0500, ElecGas 4.40

Which have the lowest?

OtherMining 0.9, Fishing 1.25, Most of the service industries 1.9

3.7. What is the value at delivered prices of household consumption for imported "FoodProds" (delivered price = basic + margins) (Header BSMR)? *35,121*

What is the value at basic prices of household consumption of domestically-produced commodity " FoodProds " (including margins)? *313,620*

3.8. What is the total value of capital rental for industry "Construction" in Indonesia? (Header 1CAP) *97,408*

Which region has the largest share of the total value of capital rental for industry "Construction".

North West Java (36.4%)

3.9. What is the value of investment demand for commodity "TranspEquip" by the industry "Coal" in Indonesia? (Header 2PUR) *621*

Which province has the highest demand for “TranspEquip” by their industry “Coal”?

East Kalimantan(580)

4. Consequences of the data

You will also need to look at consequences of the data (for example, totals and shares calculated from the data which is read in).

The main TABLO Input file TERM.TAB can be used for this purpose. As well as solving the model, it writes out various useful data calculated from the original data to a SUMMARY file. You can use GEMSIM to produce this file without actually performing a simulation.

Open TERM.TAB and search for "summary". You will see that some Coefficients are written to a SUMMARY file. To produce the SUMMARY file proceed as follows.

Open and examine the file NoSim.CMF. In the DOS box type the following command:

```
tabmate nosim.cmf
```

It is a special CMF file which just processes the model data, without doing a simulation. Hence, no closure or shocks are needed. It will read all the data, calculate all coefficients, and write the summary data. Now type:

```
tablo -pgs term
```

To run the command file type:

```
gemsim -cmf nosim.cmf
```

The program should complete without error. Once the run has finished, type into the DOS box:

```
viewhar termsum.har
```

to examine the SUMMARY file.

If you want to arrange the headers alphabetically, click on

File | Options and activate **Sort headers alphabetically**.

Most of the following questions can be answered by examining TERMSUM.HAR. Each data item there has a descriptive "long name". As a guide, you might in each case look first in the TAB file to find

- the correct name of the COEFFICIENT.
- how each item was calculated (and whether it is really what you want to find out).
- the Header used in writing this COEFFICIENT to file SUMMARY.
- then look in TERMSUM.HAR in ViewHAR to find the Header and double-click on it to see the values.

ViewHAR options: You can get ViewHAR to show the name of the Coefficient by selecting

File | Options

and then clicking on the box **Show coefficient names**.

Another good idea is to select

File | Use advanced editing menu rather than

Use simplified, read-only menu.

4.1. The COEFFICIENT used for the total labour bill by industry and province is LAB_O(i,d). What is the total labour costs for the Cement industry? *2904*

4.2. What COEFFICIENT is used for the total cost in industry "i" (plus tax)? [Hint search in TERM.TAB for “total national industry cost plus tax”]. What is the total value of this coefficient? Under what header does this coefficient appear in TERMSUM.HAR?

NATVTOT, 5836364, NATV

4.3. What COEFFICIENT is used for the total factor input to industry "i"? What is the total primary factor payments for all industries? (PRIM) *PRIMCOST, 3131009*

What is the total value of capital rental summed over all industries? *1514466*

What is the total payments to labour summed over all industries? (LAB_O) *151032*

What is the total value of factor input to industry "Coal"? *39763*

What is the total value of factor input to industry "OilPalm"? *12231*

What is the share of each of the factor inputs in the total costs of industry "Coal"?

Land = 26%, Labour = 22% and Capital = 52%

What is the share of each of the factor inputs in the total costs of industry "OilPalm"?

Land = 6%, Labour = 82% and Capital = 12%

What is the share of labour in total factor cost for all industries? *1510132/3131009 = 48.2%*

4.4. What is the total value at purchasers prices of exports of domestically-produced commodities "EdibleOil" and "OilPalm"? [Hint: search for the header VEXP]

OilPalm = 175, EdibleOil = 49054

What COEFFICIENT is this? *EXPORT*

What set elements do you need to look up? *EXPORT("EdibleOil") summed over all regions and EXPORT("OilPalm") summed over all regions.*

What header on file SUMMARY *VEXP*

Which region exports the most "Edibleoil" and "OilPalm"? *EXPORT("EdibleOil") = East Java=30%*

EXPORT("OilPalm") = West Sumatra = 58%

What commodity has the greatest share nationally? [Hint: use the "column" share" feature in ViewHAR]

VEXP OilGas 10.9% followed by PetrolRefin at 10.6%

Which province has the highest share of agricultural exports? [Hint: sum the commodity exports of the first five commodities for each region]. *Bali 59.12%*

4.5. What is the difference between the Coefficients GDPINCSUM and GDPEXPSUM in TERMSUM.HAR?

GDPEXPSUM summarises GDP from the expenditure side while GDPINCSUM summarises GDP from the income side

What is the value of household expenditure in the provincial domestic product (PDP) of NusaTeng? *42518*

What proportion is it of the PDP? (Use "Row" in the top left drop down box). *65.5%*

What proportion is NusaTeng's PDP of Indonesia's GDP? (Use "Col" in the top left drop down box) . *2%*

Which region PDP is the largest contributor of Indonesia's GDP? *North West Java (NthWestJava) at 35%*

4.6. Which header in TERMSUM.HAR tells us about Sales structure? *VMAINDEST*

Look at the Sales shares. Which industry sells the highest fraction of its output to investment? Set the combo boxes to "All COM", "Sum REG" and "All MAINDEST". After that, which is second highest? Explain.

Construction 90% Machines 6%. These inputs are used in capital creation.

What is the main destination of the coal produced in Indonesia? *Exported 72.5%*

What is the main destination of OilPalm produced in Indonesia? *Intermediate use 59%*

4.7. Look at the header CSTM in TERMSUM.HAR which tells us about Cost structure. What is unusual about the cost structure of "RealEstateDo"? [Hint: Set the combo boxes to “All IND”, “All COSTCAT” and “Sum DST”. *High capital share of 93.3%*

What does the production of this industry represent? *Rents and “imputed” rents*

Which three industries have the highest labour costs? *Crops (82%), Paddy (71.2%) and GovServices (69%)*

4.8. The header MSHR in TERMSUM.HAR shows the import penetration for each commodity: ratio of (sales of imported good)/(total domestic sales of domestic+imported good) . Look at the Formula for IMPSHR in the TERM.TAB file. Which two commodities have the highest import penetration in East Sumatra? *WaterTrans = 58.26 %, BasicMetals (53.56%)*

4.9. Use the header VMNU to see the ratio of export sales to total sales of domestically-produced commodity c. Which commodities export more than 50 percent of the value produced? [Hint: Set the combo boxes to “All COM”, “1dom”, “All MAINUSR” and “Sum DST”].

Coal (74.3%), OilGas (50.4%), WaterTrans (55.2%).

Which commodities are not really exported? *Paddy, ElecGas, Water, Construction.*

GDP questions

4.10. Which 3 headers in TERMSUM.HAR show values of GDP from the income and expenditure sides? *GNSM, GESM, NGSM*

4.11. Are the values of Nominal GDP from the income and expenditure sides equal? *Yes at 3196081*

4.12. Nominal national GDP from the expenditure side is often given by the formula

GDP = C + I + G + X - M

where C is Consumption, I is Investment, G is Government, X is Exports, M is Imports.

What COEFFICIENT contains these values in TERMSUM.Har? *NATGDPEXPSUM at header NGSM*

Find the values of all these quantities and determine each value’s share in GDP.

(Hint: Textbook definitions of GDP often ignore inventories.)

Fill in the following list of values:

	Value (in the Summary file)	Share of GDP (“col” share)
C	*2086342*	*65.3%*
I	*693007*	*21.7%*
G	*224975*	*7%*
X	*977176*	*30.6%*
-M	-*808618*	-*25.3%*
GDP	*3196081*	100%

4.13. Examine coefficient GDPEXPSUM at header GESM. This shows expenditure side GDP by region. New columns REXPORTS and RIMPORTS are needed to account for trade within Indonesia. [EXP and IMPORTS include only international exports]. Choose the *Row shares* option to see shares of each category in regional GDP.

What is the share of EXPorts in West Sumatra GDP? *33.3%*

What is the share of EXPorts in Bali GDP? *0.2%*

4.14. Nominal GDP from the income side is given by the formula

GDPINC = PRIM + TAX

What do PRIM and TAX represent? *Primary factors and all the commodity and production taxes.*

Go to TERM.TAB and search for “GNSM”. You will find the formula for the calculation of GDP from the income side to see what “PRIM” and “TAX” mean. Fill in the following list of values from TERMSum.har (choose "Sum DST"):

	Value (in the Summary file)	Share of GDP (“col” share)
LAND	*106411*	*3.3%*
LABOUR	*1510132*	*47.2%*
CAPITAL	*1514466*	*47.4%*
PRODTAX	*-50894*	*-1.6%*
COMTAX	*115967*	*3.6%*
GDPINC	*3196081*	100%

Share of PRIM in GDP (PRIM / GDPINC) = *98%*

Still viewing Nominal GDP from the income side at header, GNSM, choose "All DST" and the Col shares option to see shares of regional GDP within the nation. Look at the total column.
What is NthWestJava share in national GDP ? *34.9%*
What is Maluku share ? *0.2%*